LendWise AI-001 — Technical Documentation Skeleton

Document status: Draft template — evidence incomplete
Version: 0.1
System ID: AI-001 — Credit Scoring
Organization: LendWise Financial Services Ltd. (fictional)
Related artifact: `Gap_Assessment.csv`

Document Control

Field	Value

System name	LendWise AI-001 Credit Scoring
Provider identity	To be confirmed
System version	To be documented
Intended deployment	European Union and India — exact deployment scope to be confirmed
Intended purpose	Generate a credit-risk score to support human lending decisions
Document owner	Technical Owner — role holder to be assigned
Review status	Draft; not approved

Last reviewed	To be recorded

1. System Description and Intended Purpose

Describe the system's purpose, architecture, interfaces, model version, intended users, affected persons, deployment environment, and interactions with other systems.

Known scenario facts: AI-001 generates a credit-risk score. Humans are intended to make lending decisions.

Evidence required: Approved system description, architecture diagram, interface inventory, version history, intended-purpose statement, and documented exclusions.

Open gaps: Actual production architecture, version, provider identity, and deployment configuration have not been independently verified.

2. Provider and Responsible Parties

Record the legal entity responsible for the system and the applicable roles of provider, deployer, importer, distributor, or other relevant operators.

Evidence required: Legal-entity details, system ownership record, contracts, responsibility allocation, and relevant registration details.

Open gaps: The applicable legal roles and formal ownership arrangements require confirmation.

3. System Design and Development

Document the development process, model architecture, key design choices, training methodology, software dependencies, infrastructure, and version-control arrangements.

Evidence required: Technical design, model configuration, development records, software and dependency inventory, and change history.

Open gaps: Complete system-level technical documentation has not been verified.

4. Data Requirements and Data Governance

Describe training, validation, and testing data as applicable; data sources and provenance; collection and preparation; quality criteria; representativeness; known limitations; and relevant bias-mitigation measures.

Existing evidence reported: Dataset-validation notes and the model audit notebook using the public UCI German Credit dataset.

Evidence required: Dataset documentation, lineage, suitability assessment, quality checks, label definitions, and documented bias analysis.

Open gaps: Proxy-variable risks, label reliability, representativeness, and other data-suitability questions remain under review.

Important limitation: The public UCI dataset is a training and portfolio dataset. Its analysis does not establish the quality or representativeness of any real LendWise production dataset.

5. Risk Management

Describe the risk-management process across the AI lifecycle, including identification, estimation, mitigation, monitoring, and residual-risk review.

Existing evidence reported: NIST AI RMF assessment, AI impact assessment, and consolidated AI-001 remediation tracker.

Evidence required: Traceable risk register, control mappings, risk-treatment decisions, monitoring criteria, and authorized residual-risk decisions.

Open gaps: Outstanding remediation and operational control evidence must be resolved as applicable.

6. Model Performance and Validation

Document intended performance measures, validation methodology, test datasets, baseline results, acceptance criteria, limitations, and independent validation.

Existing evidence reported: Model card, audit report, and notebook documenting the baseline logistic regression model.

Reported baseline results: Accuracy 0.775; precision for the bad-credit class 0.6364; recall 0.5833; F1 score 0.6087.

Evidence required: Reproducible validation records, approved acceptance criteria, documented limitations, and appropriate independent review.

Open gaps: The reported baseline is an educational experiment and does not demonstrate production performance or regulatory conformity.

7. Fairness and Bias Assessment

Document relevant group-level analyses, proxy-variable risks, limitations in subgroup sample sizes, and remediation decisions.

Existing evidence reported: Age-group and housing-group analyses, foreign-worker feature-removal comparison, and associated notes.

Evidence required: Justified group selection, appropriate metrics, uncertainty analysis, proxy-variable assessment, and documented mitigation decisions.

Open gaps: The existing audit identified unresolved fairness and data-quality questions. Small subgroup sizes limit reliable conclusions.

8. Human Oversight

Describe how human decision-makers interpret model outputs, exercise authority, override recommendations, escalate concerns, and avoid overreliance.

Existing evidence reported: Human oversight matrix and F-004 control-evidence and closure matrix.

Evidence required: Approved oversight procedures, assigned decision rights, competency records, operational test evidence, and review logs.

Open gaps: Implementation, operational effectiveness, and independent closure evidence remain outstanding.

9. Logging and Record-Keeping

Document required events, log content, timestamps, retention, integrity, access controls, monitoring, and retrieval arrangements.

Existing evidence reported: F-004 audit logging test plan and closure matrix.

Evidence required: Implemented logging design, retention requirements, access-control evidence, operational test results, and sample records.

Open gaps: Operational implementation and successful test results have not been verified.

10. Transparency and Instructions for Deployers

Document system capabilities, limitations, intended purpose, performance information, foreseeable risks, and instructions required by relevant users and deployers.

Existing evidence reported: Model card, audit report, and governance artifacts.

Evidence required: A completed review against applicable transparency and deployer-information requirements, supported by approved documentation.

Open gaps: Existing artifacts have not been demonstrated to cover every applicable requirement.

11. Accuracy, Robustness, and Cybersecurity

Describe applicable accuracy targets, robustness measures, resilience requirements, cybersecurity controls, and testing methods.

Existing evidence reported: Baseline model evaluation and AI-001 remediation documentation.

Evidence required: Defined acceptance criteria, robustness and security test results, monitoring procedures, and documented remediation.

Open gaps: Baseline performance metrics alone do not demonstrate comprehensive robustness or cybersecurity.

12. Quality Management and Change Control

Document applicable quality-management arrangements, version control, release approval, change assessment, supplier dependencies, and post-deployment review.

Existing evidence reported: Governance charter, RACI draft, intake form, and remediation tracker.

Evidence required: Approved procedures, formal role assignments, release records, change history, and evidence of operation.

Open gaps: Formal approval and implementation of the operating model remain unverified.

13. Conformity Assessment and Registration

Record the applicable conformity-assessment route, supporting evidence, required declarations, registration obligations, and responsible parties, where applicable.

Evidence required: Documented legal analysis, completed applicable assessment records, and registration or declaration evidence as required.

Open gaps: The conformity route and completion status have not been established.

Important: This skeleton is not a declaration of conformity and does not establish that registration or assessment has been completed.

14. Post-Market Monitoring and Corrective Action

Describe applicable monitoring, incident handling, complaints, corrective action, escalation, and relevant reporting arrangements.

Existing evidence reported: AI-001 remediation tracker and governance documentation.

Evidence required: Approved monitoring plan, escalation and corrective-action procedures, assigned owners, and operating evidence.

Open gaps: Operational monitoring and the required supporting evidence remain to be established or verified.

15. Evidence Index

For every requirement, record:

- Evidence ID.
- Document or record name.
- Version and date.
- Requirement or finding addressed.
- Evidence owner.
- Repository location.
- Review status.
- Limitations or outstanding actions.

A document's existence does not by itself demonstrate that the associated requirement has been satisfied.

16. Open Issues and Approval Status

The following remain outstanding as applicable:

- Final legal classification and scope review.
- Completion of the conformity gap assessment.
- Resolution and validation of required remediation.
- Complete technical documentation and evidence index.
- Implementation and testing evidence for required controls.
- Independent review and authorized approval.

Current decision: AI-001 remains Not Approved — Remediation Required.

Document status: Draft template; incomplete evidence; not approved for production.

17. Change History

Version	Change	Status
0.1	Initial technical documentation skeleton	Draft